WORKED SOLUTION

STEP II 1996, Question 11

Mechanics • Simple Harmonic Motion

STEP II 1996, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q11

The key idea

Work with the stretch of the string, which is the particle's position *relative to the lift*. Write Newton's second law for the particle in the ground frame and subtract the lift's known acceleration: what's left is simple harmonic motion about a shifted centre. The string stays taut as long as the stretch stays positive, and an SHM solution tells you exactly how low the stretch can go. When the lift stops accelerating, only the centre of the oscillation moves; the particle's position and velocity relative to the lift carry straight over, and they give the new amplitude.

Solution

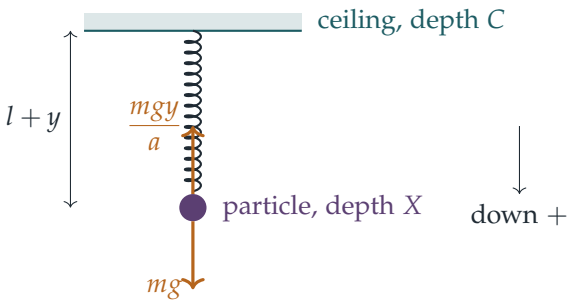
We model the particle as having mass m and the string as light and elastic, with modulus λ, obeying Hooke's law while it is taut. Throughout, y is the amount by which the string's length exceeds its natural length l, so $y = a$ at the start. This is the only reading that fits the stated differential equation. If y were measured from the old equilibrium length $l + a$, the right-hand side would be $-f$.

The equation of motion

In the initial equilibrium the tension $\lambda a/l$ balances the weight, so

$$\frac{\lambda a}{l} = mg \quad \Longrightarrow \quad \frac{\lambda}{l} = \frac{mg}{a},$$

and when the extension is $y > 0$ the tension is mgy/a.



Measure depths downwards from a point fixed to the ground: the ceiling is at depth C and the particle at depth X, so $X - C = l + y$. The lift descends with acceleration f, so $\ddot{C} = f$. Resolving vertically (downwards) for the particle, while the string is taut,

$$m\ddot{X} = mg - \frac{mgy}{a}.$$

Since $y = X - C - l$, we have $\ddot{y} = \ddot{X} - \ddot{C} = \ddot{X} - f$, and therefore

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \frac{g}{a}y = g - f,$$

as required.

The string stays taut; amplitude af/g

Put $\omega^2 = g/a$. The constant particular solution is $y = (g-f)/\omega^2 = a(g-f)/g$, so the general solution is

$$y = a - \frac{af}{g} + A\cos\omega t + B\sin\omega t.$$

At $t = 0$ the particle is in its old equilibrium and moving with the lift, so $y(0) = a$ and $\dot{y}(0) = 0$. These give $A = af/g$ and $B = 0$:

$$y = a - \frac{af}{g} + \frac{af}{g}\cos\omega t.$$

This is SHM about $y = a - af/g$ with amplitude af/g. The particle's position relative to the lift is $l + y$, so the particle oscillates relative to the lift with amplitude af/g.

Ask yourself We used Hooke's law, which assumes the string is taut. Isn't that circular?

No. We assume the string is taut and solve, then check that the solution never leaves the region where the assumption holds. The smallest value of y is

$$y_{\min} = a - \frac{2af}{g} = a\left(1 - \frac{2f}{g}\right) > 0,$$

because $f < g/2$. So y stays positive, the string is always taut and the solution is self-consistent. This is exactly where the condition $f < g/2$ is needed.

After the lift stops accelerating

For $t > T$ we have $\ddot{C} = 0$, and the same argument gives

$$\ddot{y} + \omega^2 y = g,$$

which is SHM about $y = a$, the natural equilibrium extension. Write $z = y - a$, so that $\ddot{z} + \omega^2 z = 0$.

Ask yourself What are the starting conditions for this new motion?

The lift's acceleration jumps at $t = T$, but its velocity $\dot{C}$ doesn't, and nor do the particle's position and velocity. So $y = X - C - l$ and $\dot{y} = \dot{X} - \dot{C}$ are continuous at $t = T$. From the first stage, at $t = T$,

$$z = \frac{af}{g}(\cos \omega T - 1), \qquad \dot{z} = -\frac{af}{g}\,\omega \sin \omega T.$$

For SHM $\ddot{z} = -\omega^2 z$ the amplitude R satisfies $R^2 = z^2 + \dot{z}^2/\omega^2$ at any instant (this is energy conservation, or the identity $\cos^2 + \sin^2 = 1$ applied to $z = R\cos(\omega t + \varepsilon)$). Hence

$$\begin{aligned} R^2 &= \frac{a^2 f^2}{g^2}\left[(1 - \cos \omega T)^2 + \sin^2 \omega T\right] \\ &= \frac{a^2 f^2}{g^2}\,(2 - 2\cos \omega T) \\ &= \frac{a^2 f^2}{g^2} \cdot 4 \sin^2 \tfrac{1}{2}\omega T, \end{aligned}$$

using $1 - \cos\theta = 2\sin^2 \frac{1}{2}\theta$. Therefore the new amplitude is

$$R = \frac{2af}{g}\,|\sin \tfrac{1}{2}\omega T|,$$

as required.

The string never becomes slack: the least extension is now

$$a - R \;\geq\; a - \frac{2af}{g} = a\left(1 - \frac{2f}{g}\right) > 0,$$

again because $f < g/2$. So $y > 0$ throughout and the equation of motion stays valid.

Checks

- If T is a whole number of periods, $\omega T = 2k\pi$, the particle is back at $y = a$ and at rest relative to the lift just when the lift stops accelerating. The formula gives amplitude 0: the particle hangs still, as it should.

- If ωT is an odd multiple of π, the particle is at its highest point, $y = a - 2af/g$, and momentarily at rest relative to the lift. The new centre is $y = a$, so the amplitude is $2af/g$, which is the maximum value of the formula.
- Numerically, we stepped the full motion in the ground frame (lift and particle, with tension switched off if the string went slack) for random a, $f < g/2$ and T. For example, with $a = 0.546$, $f/g = 0.337$ and $\omega T = 15.27$, the amplitudes came out as 0.18380 and 0.35864, matching af/g and $\frac{2af}{g}|\sin\frac{1}{2}\omega T|$ to six figures, and the extension never fell below $a - 2af/g$.

Common slips

- Measuring y from the old equilibrium length $l + a$. Then the right-hand side would be $-f$, not $g - f$.
- Restarting the second stage with $\dot{y} = 0$. The particle is generally moving relative to the lift at $t = T$, and that velocity contributes to the amplitude.
- Getting the sign of f wrong. A lift accelerating *downwards* reduces the effective gravity to $g - f$, so the centre of oscillation rises.

Beyond STEP

Subtracting the lift's acceleration is the same as working in an accelerating frame, where a uniform acceleration f acts like an extra gravitational field $-f$. That idea leads to fictitious forces and, eventually, to Einstein's equivalence principle. The second stage is a simple case of an oscillator driven by a step change in force. Its response depends on when the step happens relative to the oscillation, and the general tool for this is Green's functions (Duhamel's principle). Kleppner and Kolenkow, *An Introduction to Mechanics*, treats non-inertial frames and oscillators in this spirit.

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